

Autodesk® Scaleform®

Scaleform Best Practices Guide

Scaleform 4.0

: Matthew Doyle
: 3.0
: 2011 4 25

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Autodesk® Scaleform® 4.3

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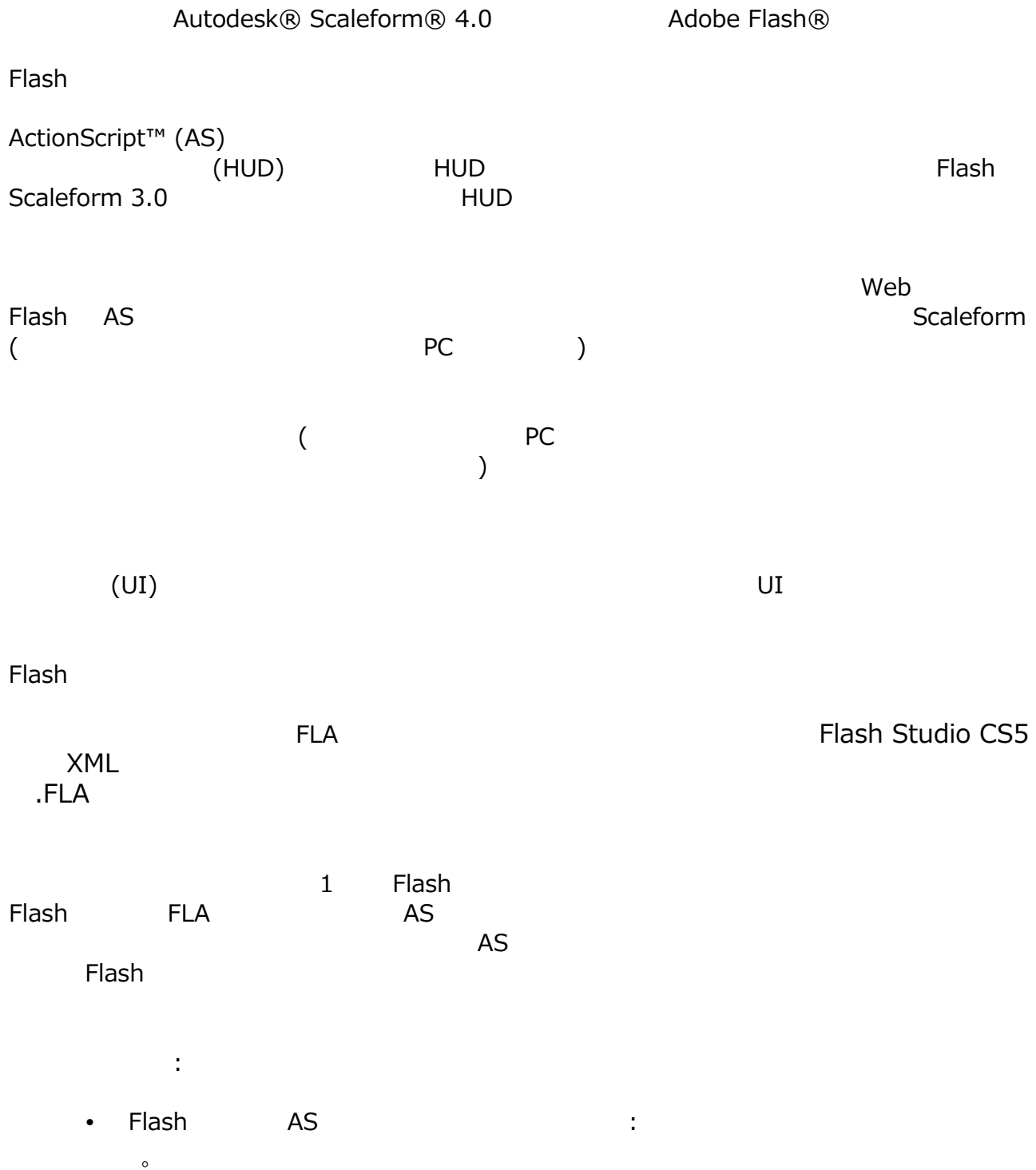
(301) 446-3200

Fax

(301) 446-3199

1		1
1.1		3
1.2		3
2		4
2.1		4
2.2		4
2.3		5
2.3.1	vs	5
2.3.2		5
2.3.3		7
2.3.4		9
2.3.5		11
3	ActionScript	13
3.1	ActionsScript	13
3.1.1		16
3.1.2		16
3.1.3	/	16
3.2	ActionScript 3	17
3.2.1		17
3.3	Advance	18
3.4	onEnterFrame and Event.ENTER_FRAME	18
3.5	ActionScript 2	19
3.5.1	onClipEvent on	19
3.5.2	Var	20
3.5.3		21
3.5.4		22
3.5.5		22
4	HUD	24
4.1	SWF	24
4.2	SWF 1	25
4.3	1	26
4.4	1 ().....	26
4.5	Flash HUD	27
5		29
5.1	Flash	29
5.2		29
6		30

1



- FLA AS :
- AS

- :
-

- Flash AS :
- Flash

Flash

Flash AS

Flash :

-
-

ActionScript

1

Flash

Flash

1.1

- 1
- UI (Flash [])

Scaleform 3.3
Scaleform Common Lightweight Interface Kit (CLIK™)

1.2

-
- The diagram illustrates the relationships between various game development components and their associated file formats or APIs. The components are arranged in a grid-like structure with labels like MMORPG, HUD, AS3, SWF, C++ API, Flash, AS, API, AS2, Scaleform 3.3, CPU CD, DVD, UI, HUD, ActionScript, and file size ranges like 400K-1.5M, 700K-1.5M, 800K-4M, and 2M-10M+.
- | Component | File Format / API | File Size Range |
|---------------|-------------------|-----------------|
| MMORPG | Flash | 400K-1.5M |
| HUD | Flash | 400K-1.5M |
| AS3 | SWF | 700K-1.5M |
| AS2 | AS | 800K-4M |
| Scaleform 3.3 | API | 2M-10M+ |
| CPU CD | Flash | 800K-4M |
| DVD | Flash | 800K-4M |
| UI | Flash | 800K-4M |
| HUD | Flash | 800K-4M |
| ActionScript | Flash | 800K-4M |

2

Flash

2.1

GFx DP Flash

DP GFx

Display DP

DP DP

DP DP

HUD F2 Summary

Scaleform Player HUD

⋮

- ## 1. DP

DP

- 2.

DP

- 3.

DP

- 4.

53.DP ~~am p i n g t o r k i n y e~~ @PAM00

✓ ~
DPP

2.

3. AS2) and `visible=false` rather than `alpha=0` (in AS3) `_alpha=0` `_visible=false`(in "stop()")

2.3

2.3.1 vs

Flash

Scaleform

2.3.2

(
(
)
)

•

•

O

Flash

0

Flash

false

•

Player

O

)

•

O

○

•

•

○

○

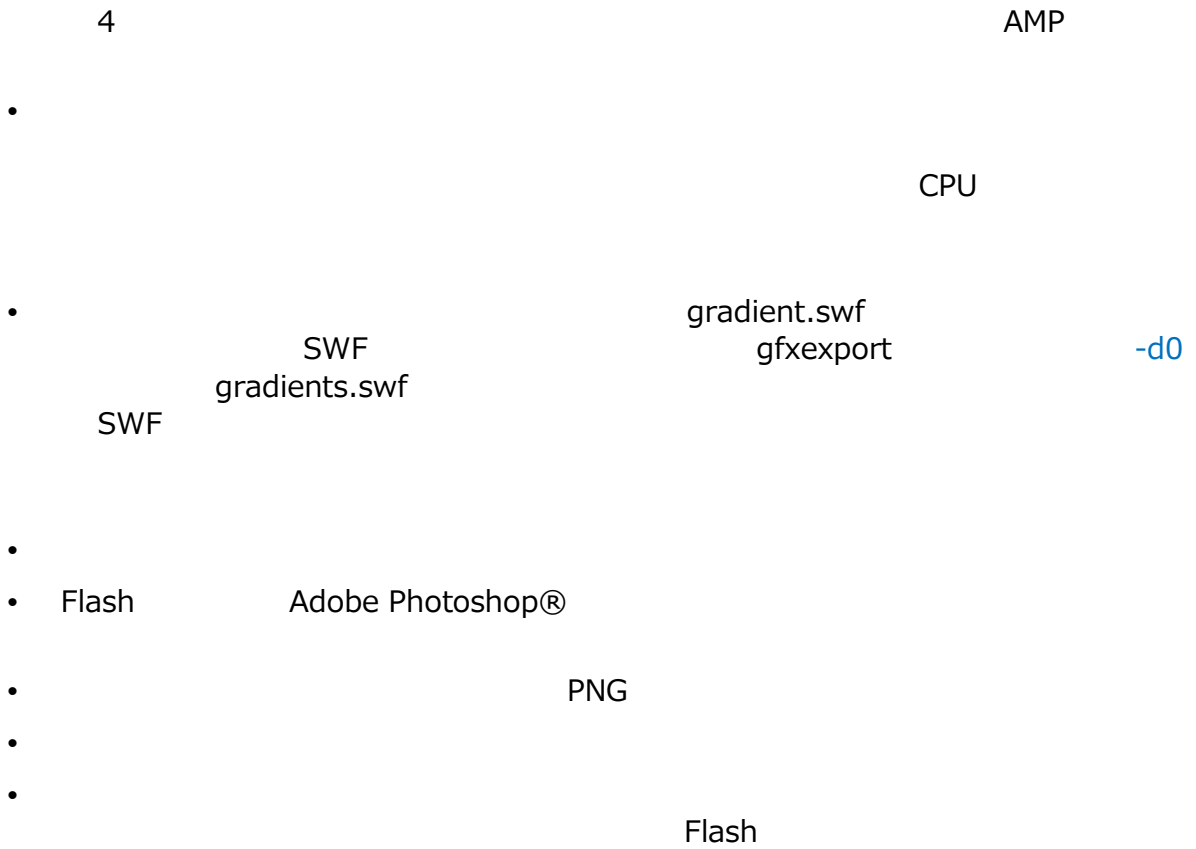
◦			2
•	Flash API	API	1
•			
	Flash Studio		
	Scaleform	EdgeAA	
•			
•			
•	EdgeAA		1
	/		
	/		

2.3.3

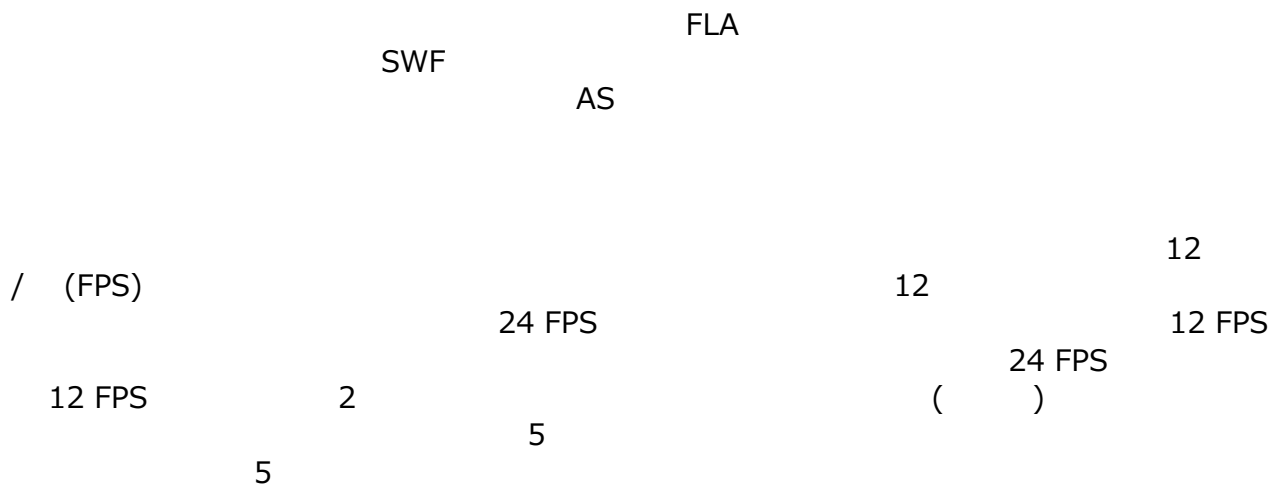
Flash 10

SWF `_alpha`(in AS2) or `alpha` (in AS3) 0
 `_visible`(in AS2) or `visible` (in AS3)
 false Scaleform `_alpha/alpha` 0 CPU
 CPU ActionScript false SWF
 `_visible/visible` false
 :

- 2 /
 16x32 256x128 1024x1024 512x32
- JPEG
- gfxexport
 DXT SWF GFx
 DXT gfxexport
 `-qp` `-qh` DDS ()
- Edge AA
- ActionScript /
- SWF/GFX . SWF SWF
- 1024x1024 JPEG 4 MB
- UI 4 () gfxexport `-i DDS`



2.3.4



: `onEnterFrame`(in AS2) or `Event.ENTER_FRAME` (in AS3)

`onEnterFrame/Event.ENTER_FRAME`

`setInterval`

`onEnterFrame/EVENT.ENTER_FRAME`

`setInterval`

30 40 FPS
CPU

Flash

UI

:

•

•

/

/ `_visible/visible = true/false`

•

◦

1

◦

Flash

◦ Tween
Scaleform CLIK Tween (gfx.motion.Tween in AS2 or
scaleform.clik.motion.Tween in AS3)
Tween

Flash

•

60

Flash

Flash

60 FPS

•

•

•

(

)

•

-

-

CPU

(
()

CPU
)

4

/

-

()

2.3.5

-

gfxexport

(48)

SlotHeight

DP
)

(

DP

-

DP 1

-

1

30 FPS

-

("TextField.variable")

-

"htmlText"

HTML

-

gfxexport

-fc -fcl -fcm

(

)

"Font and Text Configuration Overview"

- fontlib
("Font and Text Configuration Overview"
)
- TextField 1 1
1 DP
- 1 (1)
- ()
-
- DrawText API DrawText API
Flash Scaleform C++
Flash UI DrawText API C++ 1

3 ActionScript

ActionScript

AS

AS

3.1 *ActionsScript*

AS

- SWF Flash 8
- AS

AS

[attachMovie](#)

FLA

- AS
-
-
- "if"

- [on\(\)](#) [onClipEvent\(\)](#) [onEnterFrame](#)
[onPress](#)

- AS

Flash AS

([onClipEvent](#) [on](#))

([gotoAndPlay](#) [play](#) [stop](#))

- , [gotoAndPlay/gotoAndStop](#) ,

- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
 gotoAndPlay/gotoAndStop
- Gfx::Movie::SetVariableArray
 1
 1
 C++ C++ AS 1
 1
- 1
 Scaleform
 Scaleform
- Object
 ◦ Object
- eval() 1
- myArr.length : Array.length
- ```
var fontArr:Array = TextField.getFontList();
var arrayLen:Number = fontArr.length;
for (var i:Number = 0; i < arrayLen; i++) {
 trace(fontArr[i]);
}
```

```

:
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
 trace(fontArr[i]);
}

```

- ( null )
- removeListener()
- - AS VM
  - com.xxx.yyy.aaa.bbb VM if AS
  - aaa bbb com xxx yyy if
  - com xxx xxx yyy
  - Flash SWF
  - c58923409876.functionName()
- - AS SWF SWF
- AS
- - SWF [ ] trace()
  - [Flash] [Trace
  - trace
- 
- - 1 SWF AS ( : foo.bar.CustomClass)
  - SWF SWF

delete

:

delete foo.bar.CustomClass

- 100% ( )
- onEnterFrames
- ( ) Scaleform

### 3.1.1

- 
- 
- 
- 
- 

### 3.1.2

- 
- with

### 3.1.3 /

- 
- "var" ActionScript "var"
-

- 
- ```

graph LR
    Node1["SWF  
null"] --> Node2["AS  
getter"]
    Node2 --> Node3["getter  
setter"]
    Node3 --> Node4["null"]
  
```

3.2 *ActionScript 3*

AS3

3.2.1

- Object
- Array
- Vector
- Vector<*>
- Vector<String>
- Object
- AS2
- AS3

3.3 Advance

Advance

6 :

1. AS `onEnterFrame/Event.ENTER_FRAME`
2. `UI` `Invoke`
3. `(AS3)` `true` `stop()` `(_visible(AS2) or visible`
`)` `Advance` `1` `1`
4. `_global.noInvisibleAdvance(AS2) or scaleform.gfx.Extensions.noInvisibleAdvance`
`(AS3)` `Advance` `"true"` `(`
`)` `Advance` `Flash`
`Flash` `(`
`)` `_global.gfxExtensions(AS2) or`
`scaleform.gfx.Extensions.enabled (AS3)` `true` `Scaleform`
5. `Advance`
- 6.

3.4 *onEnterFrame and Event.ENTER_FRAME*

`onEnterFrame(AS2) or Event.ENTER_FRAME (AS3)`

`onEnterFrame/Event.ENTER_FRAME`
`setInterval` `setInterval` `setTimeout`
`setInterval` :

- clearInterval
- setInterval setTimeout onEnterFrame/Event.ENTER_FRAME
onEnterFrame/Event.ENTER_FRAME

onEnterFrame delete :

```
delete this.onEnterFrame;
delete mc.onEnterFrame;
```

onEnterFrame null undefined (this.onEnterFrame = null;)
onEnterFrame onEnterFrame
Scaleform

3.5 ActionScript 2

3.5.1 onClipEvent on

onClipEvent() on() onEnterFrame onPress
:

-
- onClipEvent on
this _global arguments super
256

onLoad

onEnterFrame : onClipEvent(construct)

```
onClipEvent(construct)
{
    this.onLoad = function()
    {
        //
    }
}
```

onClipEvent(load)

3.5.2 Var

var

AS

var

AS

2

:

```
var i = 1000;
countIt = function()
{
    num = 0;
    for(j=0; j<i; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}
```

:

```
var i = 1000;
countIt = function()
{
    var num = 0;
    var ii = i;
    for(var j=0; j<ii; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}
```


3.5.3

) (var

```
function foo(var obj:Object)
{
    for (var i = 0; i < obj.size; i++)
    {
        obj.value[i] = obj.num1 * obj.num2;
    }
}
```

```
function foo(var obj:Object)
{
    var sz = obj.size;
    var n1 = obj.num1;
    var n2 = obj.num1;
    for (var i = 0; i < sz; i++)
    {
        obj.value[i] = n1*n2;
    }
}
```

```
var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);
```

```
var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
    // ;
}
```

3.5.4

:

```
mc.ch1.hc3.djf3.jd9._x = 233;  
mc.ch1.hc3.djf3._x = 455;
```

:

```
var djf3 = mc.ch1.hc3.djf3;  
djf3._x = 455;  
  
var jd9 = djf3.jd9;  
jd9._x = 223;
```

3.5.5

C

:

```
this[_global.mynames[queue]][_global.slots[i]].gosplash.text =  
_global.MyStrings[queue];
```

:

```
var _splqueue = this[_global.mynames[queue]];  
var _splstring = _global.MyStrings[queue];  
var slot_i = _global.slots[i];  
_splqueue[slot_i].gosplash.text = _splstring;
```

:

```
for(i=0; i<3; i++)  
{  
    this[_global.mynames[queue]][_global.slots[i]].gosplash.text =  
        _global.MyStrings[queue];  
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =  
        _global.MyStrings[queue];  
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
```

```
0x000000;  
}
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
    var slot_i = _global.slots[i];
    _splqueue[slot_i].gosplash.text = _splstring;
    _splqueue[slot_i].gosplash2.text = splstring;
    _splqueue[slot_i].gosplash2.textColor = 0x000000;
}
```

:

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
    var slot_i = _global.slots[i];
    var elem = _splqueue[slot_i];
    elem.gosplash.text = _splstring;
    var gspl2 = elem.gosplash2;
    gspl2.text = splstring;
    gspl2.textColor = 0x000000;
}
```

4 HUD

(HUD)

Scaleform HUD

HUD

4.1

HUD

HUD

HUD

C++

4.1 SWF

HUD

80K

HUD

SWF

■

■

1. 別々のスレッド上で *Advance* を呼び出す機能。

Flash

HUD

Advance

()

AS

Flash

Advance

Flash

Advance

HUD

Advance

Advance

HUD

Advance

: Scaleform

Gfx::Movie

Advance

	Invoke	Display Advance		Input
--	--------	--------------------	--	-------

2. *render-to-texture* キャッシングを利用する。 HUD

HUD

4.2 SWF 1

Flash	AS loadMovie	1
-------	------------------------------	---

Display

Flash	1
-------	---

1. [_visible = false](#)(AS2) or
 visible = false (AS3) [_global.noInvisibleAdvance](#)(AS2) or
 scaleform.gfx.Extensions.noInvisibleAdvance (AS3) true Advance
 HUD 1

[_visible/visible = false](#)
 (:
 Advance
 Flash
 HUD /
) HUD / ([_global.gfxExtensions](#)(AS2) or
 Advance [scaleform.gfx.Extensions.enabled](#) (AS3) true Scaleform

2. [SetVariableArray](#) 1 Invoke

() 1

[SetVariableArray](#) ()
) 1 Invoke

1

3. HUD onEnterFrame(AS2) or
Event.ENTER_FRAME (AS3)
HUD onEnterFrame/Event.ENTER_FRAME
Advance
onEnterFrame
4. AS

tick
AS Invoke Flash

15-20 FPS 30-40 FPS

HUD
- 5.

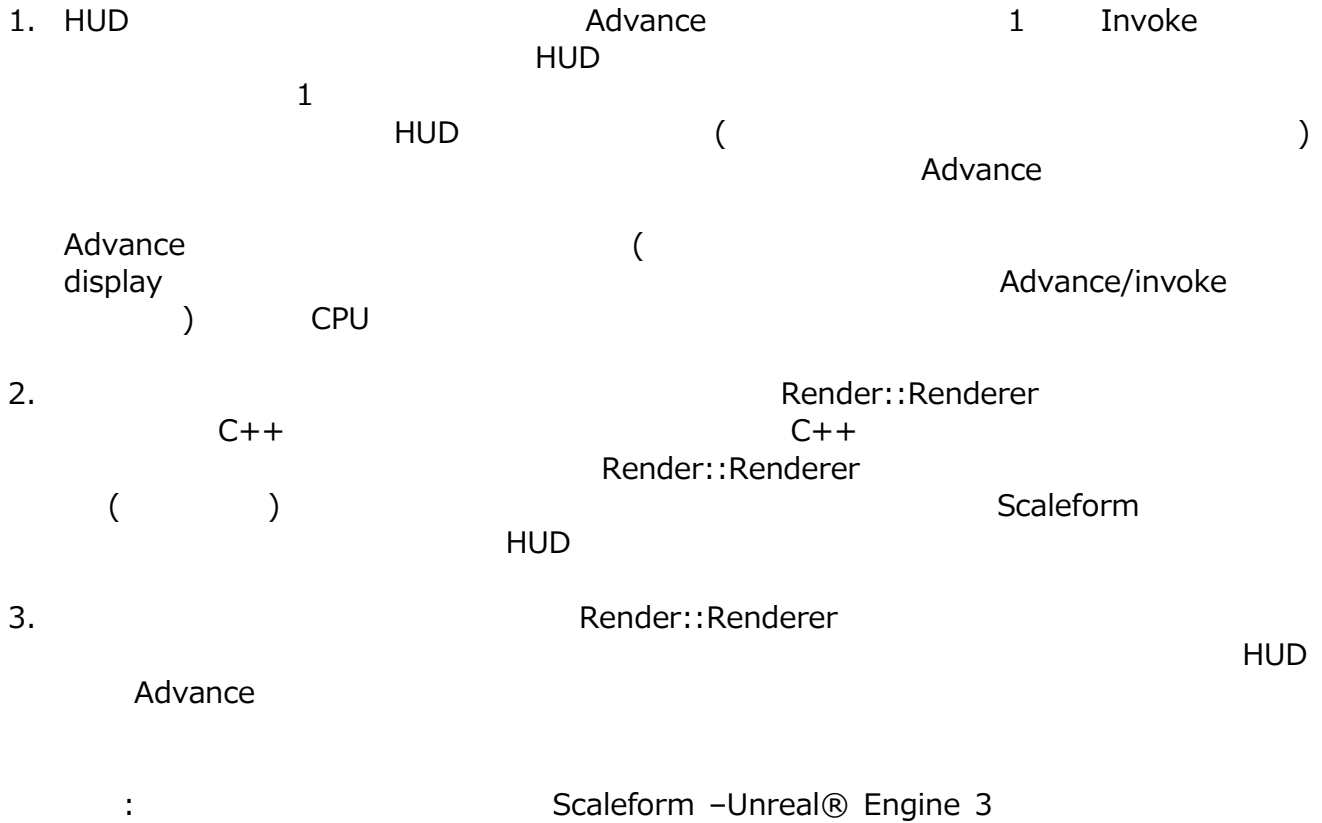
4.3 1

- Flash () Flash
- Flash C++
Flash
- 4.2
- :
1. Flash
Scaleform HUD
 2. C++ Flash
Flash
Flash 1
Render::Render
RenderString
C++
AS

4.4 1 ()

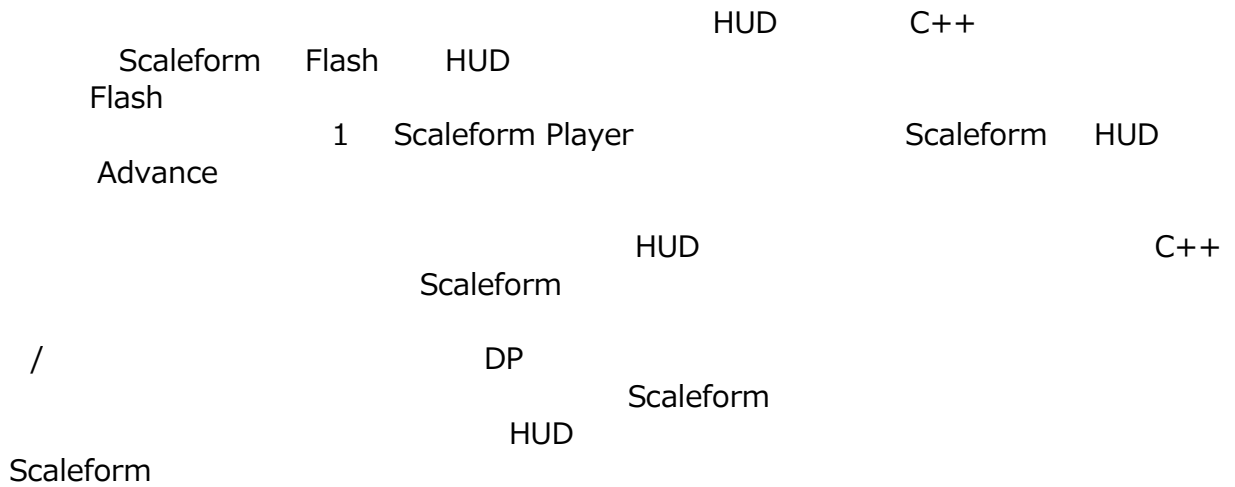
HUD
4.3

:



4.5 Flash

HUD



[Scaleform](#) / [Flash](#)
[Scaleform](#) C++
[DrawText API 1](#) [Scaleform 2](#) [HUD 1](#)
[Scaleform](#) / [Scaleform FAQ](#) ["Font & Text"](#) [Font and](#)
[Text Configuration Overview](#) / [DrawText API](#)

- HUD :
- - ["Graphics Rendering & Special Effects"](#) 1 2 FAQ
 - PC Wii™ (
 - Gfx::Movie::SetMouseCursorCount(0);
 -
 - / HUD
noInvisibleAdvance (AS2) or scaleform.gfx.Extensions.noInvisibleAdvance (AS3)
[_visible\(AS2\) or visible \(AS3\) = false](#)
 - Invoke 1 2 Invoke
()
()
 - [2.1](#)
FAQ ["Art & Assets"](#)

5

5.1 Flash

Flash

2

FLA

- FPS
- AS
- ([1] [2])

5.2

-
- Diagram illustrating a sequence of actions and AS (Action Set) labels:
- Vertical line of dots on the left.
 - Labels: 3, 1, trace(), (, AS, AS, actions, 1, AS.

6

ActionScript 2.0 Best Practices

http://www.adobe.com/devnet/flash/articles/as_bestpractices.html

Flash 8 Best Practices

http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html

Flash ActionScript 2.0 Learning Guide

http://www.adobe.com/devnet/flash/articles/actionscript_guide.html